

MD RASEL KHANDAKER

researcher@mdraselkhandaker.com | +60166694391 | Kuala Lumpur, Malaysia
researcher.mdraselkhandaker.com | github.com/mdraselkhandaker20-png

RESEARCH INTERESTS

Machine vision and colour science for low-cost, real-time fabric colour inspection; instrumental colour-difference measurement (CIELAB, CIEDE2000); IoT-integrated quality monitoring in smart textile manufacturing; deep learning for defect detection in Industry 4.0 environments.

EDUCATION

MSc in Engineering in Business Management

2023 – 2026

Multimedia University (MMU), Cyberjaya, Malaysia | CGPA: 3.43

- Thesis: Adoption of IoT-Enabled Quality Monitoring System in a Smart Factory — Fabric Colour Variation
- Focus areas: IoT adoption for fabric colour quality monitoring; survey research; quantitative data analysis

BSc in Textile Engineering

2018 – 2021

Sonargaon University, Bangladesh

RESEARCH EXPERIENCE

Research Project — Low-Cost Fabric Colour Variation Detection

2026 – Present

- Built a real-time machine vision system (Python, OpenCV) using a Logitech C270 webcam and a 5500 K LED booth to detect shade variation in dyed fabric
- Implemented CIELAB conversion and CIEDE2000, ΔE^*_{ab} and CMC(2:1) colour-difference metrics; verified CIEDE2000 against published test data (Sharma et al., 2005)
- Designed a setup and calibration procedure — five readiness checks and a baseline noise measurement — to derive a defensible pass/fail limit

MSc Research — IoT-Enabled Quality Monitoring in Smart Factory

2025 – 2026

- Conducted a quantitative cross-sectional study of 80 quality management professionals in eight Bangladeshi textile factories (five IoT-enabled, three non-IoT) using the TAM and TOE frameworks
- All six hypotheses supported ($p < 0.01$); IoT-enabled factories reported markedly lower shade mismatch (Cohen's $d = 4.11$) and were 5.32× more likely to report lower defect rates

PUBLICATIONS & PRESENTATIONS

Conference Paper (Presented)

2026

- Khandaker, M. R., & Thiagarajah, S. P. Adoption of IoT-Enabled Quality Monitoring System in a Smart Factory. 6th International Conference on Technology and Innovation Management (ICTIM 2026), DIFCON, Multimedia University, May 2026.

Manuscript (In Preparation)

2026

- Khandaker, M. R. Low-Cost Real-Time Fabric Color Variation Detection Using Computer Vision. Target: IEEE.

PROJECTS & SOFTWARE DEVELOPMENT

Fabric Colour Variation Detection Software

2026

- Desktop application (Flask, OpenCV, pywebview) with live analysis, calibration and CSV reports; browser version of the same method in JavaScript. Source: github.com/mdraselkhandaker20-png/FABRIC-COLOR-VARIATION-DETECTION

Smart Factory IoT Dashboard

2025 – 2026

- Developed an IoT-integrated monitoring dashboard for real-time fabric quality data visualisation.

Research Portfolio Website — researcher.mdraselkhandaker.com

2026

- Designed and developed a research website (HTML, CSS, JavaScript) with a live browser demo of the colour-difference method, hosted on GitHub Pages with a custom domain.

TECHNICAL SKILLS

Colour Science & Machine Vision: CIELAB, CIEDE2000, CMC(l:c), colour-difference measurement, image processing, OpenCV, NumPy

AI & Machine Learning: Computer vision, deep learning fundamentals (CNN, YOLO), model evaluation

Smart Factory & IoT: IoT-based monitoring systems, sensor integration, real-time data collection, quality control

Web, Software & Cloud: Python (Flask), JavaScript, HTML, CSS, PostgreSQL, SQLite, WebRTC, Flask-SocketIO, Git/GitHub, DigitalOcean, Neon, Cloudinary

REFERENCES

Available upon request (MSc Supervisor, Course Professors)